Aravind Karthik R

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Education

- **Vellore Institute of Technology** — Chennai, India
  M.Tech – Embedded Systems — RTOS, MCU Architecture, In-Vehicle Networking, Embedded Programming — *Jul 2025 – Present*
- **Vellore Institute of Technology** — Chennai, India
  B.Tech – Electronics & Computer Engineering, Minor in Management — *Jul 2021 – Sep 2025*

Skills

- **Languages**: Embedded C, C++, Python, ARM Assembly (Thumb, Interrupts), Verilog
- **Embedded & RTOS**: RTOS (QNX), Embedded Linux, Linux Device Drivers, Firmware Development, Hardware Bring-Up, STM32, PSoC, ESP32
- **Debugging & Tools**: Oscilloscope, Logic Analyzer, GDB, JTAG/SWD, Icarus Verilog, GTKWave, Quartus Prime, MATLAB
- **Protocols & Networking**: CAN (arbitration-aware, 500 kbps), SPI, I2C, UART, BLE, LoRa, MQTT, TCP/IP
- **Testing & Process**: Unit Testing, Integration Testing, Hardware-in-the-Loop Validation, Agile Development

Professional Experience

- **Indian Institute of Science (IISc)** — Bangalore, India
  Project Associate Intern — *Jun 2024 – Jun 2025*
  - Developed and optimized embedded firmware for real-time inference pipelines on Raspberry Pi, tuning memory footprint, interrupt latency, and peripheral driver performance under tight hardware constraints.
  - Wrote Python automation scripts for data acquisition and system validation; conducted hardware-in-the-loop functional testing across iterative development cycles, validating behavior on target hardware before each release milestone.
- **Stemtec AI and Robotics Pvt. Ltd. (V-NEST, VIT Chennai)** — Chennai, India
  Embedded Systems and IoT Intern — *Mar 2024 – Jun 2024*
  - Developed and tested C/C++ firmware for embedded measurement and control modules, implementing peripheral drivers, interrupt service routines, and real-time control logic.
  - Performed PCB schematic validation, hardware bring-up, and SMD soldering on prototype boards; used oscilloscope and logic analyzer for signal integrity verification and fault isolation.
  - Integrated BLE and LoRa wireless modules into low-power communication systems, configuring protocol stacks and validating link reliability across operating conditions.
- **Highbrow Diligence Services Pvt. Ltd.** — Chennai, India
  Embedded Systems Intern — *Aug 2023 – Dec 2023*
  - Developed embedded C firmware for a safety-critical medical device prototype; implemented multi-sensor SPI/I2C acquisition, real-time actuator control logic, and hardware fault-detection routines.
  - Wrote and executed unit test plans for individual firmware modules and integration test plans for sensor-actuator loops, verifying functional correctness on target hardware.

Projects

- **Adaptive CAN Bus Processor**
  ESP32, MCP2515, Python, Embedded Networking | Relevant: In-Vehicle Networking, CAN Protocol
  - Implemented a virtual multi-node CAN system on ESP32 with MCP2515 in loopback mode, simulating SAFETY (0x100), CONTROL (0x200), and TELEMETRY (0x300) nodes on a shared 500 kbps bus with standard ID-based arbitration priority.
  - Built dynamic adaptive scheduling with transmission intervals of 50–2000 ms based on a 1-second sliding bus load window; 150 ms backoff above 60% load reduced CONTROL interval from 350 ms to **139 ms** and TELEMETRY from 500 ms to **322 ms** under sustained contention.
  - Streamed real-time JSON telemetry at 115200 baud to a Python dashboard visualizing bus load, arbitration delay, and retry rates live.
- **NECTAR Biomorphic Hardware Accelerator**
  Verilog, RTL Design, Icarus Verilog, GTKWave | Relevant: Digital Design, Signal Processing Pipeline
  - Designed a custom 5-stage RISC-inspired pipeline in Verilog for real-time signal processing; achieved 100 MHz clock frequency and 9.6 Gbps peak data throughput with dedicated acquisition, DSP, feature extraction, classification, and feedback control stages.
  - Verified full RTL design using Icarus Verilog and GTKWave; resolved data and control hazards and validated timing closure; achieved approximately **65% reduction in processing overhead** versus interrupt-driven MCU implementations.
- **Decentralized MQTT-Based Resource Negotiation System**
  ESP8266, MQTT, Python | Relevant: Embedded Networking, Real-Time Arbitration
  - Designed a fully decentralized arbitration protocol across 4× ESP8266 nodes using priority scores from execution time, deadline urgency, and utilization rate; achieved **mean resolution under 180 ms** and **zero simultaneous grants** across 200+ contention events with no central coordinator.
- **LoRa-Based Smart Traffic Management for Emergency Vehicles**
  LoRa, GPS, Distributed Embedded | Relevant: Wireless Protocols, Real-Time Systems
  - Designed a distributed traffic control system using LoRa and GPS for real-time emergency vehicle detection; achieved **1.2 km** range, **35% reduction** in intersection wait time, and signal propagation latency under **120 ms** across nodes.